

Resolución de Ejercicios: Álgebra Relacional, Cálculo Relacional y SQL

1. Selección, proyección y unión

1.1 Encontrar los colores de los botes reservados por el marinero llamado Albert.

Álgebra Relacional:

$$\pi_{color}(\sigma_{sname='Albert'}(Sailors) \bowtie Reserves \bowtie Boats)$$

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s \in Sailors, \exists r \in Reserves, \exists b \in Boats (s.sname = 'Albert' \wedge s.sid = r.sid \wedge r.bid = b.bid \wedge t.color = b.color)\}$$

SQL:

```
1 SELECT DISTINCT b.color
2 FROM Sailors s
3 JOIN Reserves r ON s.sid = r.sid
4 JOIN Boats b ON r.bid = b.bid
5 WHERE s.sname = 'Albert';
```

1.2 Encontrar los sid de marineros que tengan rating ≥ 8 o que hayan reservado el bote 103.

Álgebra Relacional:

$$\pi_{sid}(\sigma_{rating \geq 8}(Sailors)) \cup \pi_{sid}(\sigma_{bid=103}(Reserves))$$

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s \in Sailors (t.sid = s.sid \wedge s.rating \geq 8) \vee \exists r \in Reserves (t.sid = r.sid \wedge r.bid = 103)\}$$

SQL:

```
1 SELECT sid FROM Sailors WHERE rating >= 8
2 UNION
3 SELECT sid FROM Reserves WHERE bid = 103;
```

2. Diferencia y negación

2.1 Encontrar los nombres de marineros que no han reservado un bote rojo.

Álgebra Relacional:

$$\pi_{sname}(Sailors \bowtie (\pi_{sid}(Sailors) - \pi_{sid}(Reserves \bowtie \sigma_{color='red'}(Boats))))$$

Cálculo Relacional de Tuplas:

$\{t \mid \exists s \in Sailors (t.sname = s.sname \wedge \neg \exists r \in Reserves, \exists b \in Boats (r.sid = s.sid \wedge r.bid = b.bid \wedge b.color = 'red'))\}$

SQL:

```
1 SELECT sname
2 FROM Sailors
3 WHERE sid NOT IN (
4     SELECT r.sid
5     FROM Reserves r
6     JOIN Boats b ON r.bid = b.bid
7     WHERE b.color = 'red'
8 );
```

2.2 Encontrar los sid de marineros con edad mayor a 20 que no han reservado un bote rojo.

Álgebra Relacional:

$$\pi_{sid}(\sigma_{age>20}(Sailors)) - \pi_{sid}(Reserves \bowtie \sigma_{color='red'}(Boats))$$

Cálculo Relacional de Tuplas:

$\{t \mid \exists s \in Sailors (t.sid = s.sid \wedge s.age > 20 \wedge \neg \exists r \in Reserves, \exists b \in Boats (r.sid = s.sid \wedge r.bid = b.bid \wedge b.color = 'red'))\}$

SQL con EXCEPT:

```
1 SELECT sid FROM Sailors WHERE age > 20
2 EXCEPT
3 SELECT r.sid FROM Reserves r JOIN Boats b ON r.bid = b.bid WHERE b.
   color = 'red';
```

SQL con LEFT JOIN:

```
1 SELECT s.sid
2 FROM Sailors s
3 LEFT JOIN (
4     SELECT r.sid
5     FROM Reserves r
6     JOIN Boats b ON r.bid = b.bid
7     WHERE b.color = 'red'
8 ) rojos ON s.sid = rojos.sid
9 WHERE s.age > 20 AND rojos.sid IS NULL;
```

3. Autoreferencia con renombrado (ρ)

3.1 Encontrar los nombres de marineros que han reservado al menos dos botes distintos.

Álgebra Relacional:

$$\pi_{sname}(Sailors \bowtie \pi_{r_1.sid}(\sigma_{r_1.sid=r_2.sid \wedge r_1.bid \neq r_2.bid}(\rho_{r_1}(Reserves) \times \rho_{r_2}(Reserves))))$$

Cálculo Relacional de Tuplas:

$\{t \mid \exists s \in Sailors, \exists r_1 \in Reserves, \exists r_2 \in Reserves (t.sname = s.sname \wedge r_1.sid = s.sid \wedge r_2.sid = s.sid \wedge r_1.bid \neq r_2.bid)\}$

SQL:

```

1 SELECT DISTINCT s.sname
2 FROM Sailors s
3 JOIN Reserves r1 ON s.sid = r1.sid
4 JOIN Reserves r2 ON s.sid = r2.sid
5 WHERE r1.bid <> r2.bid;

```

3.2 Encontrar los sid de marineros cuyo rating es mejor que el de algún marinero llamado Bob.

Álgebra Relacional:

$$\pi_{s_1.sid}(\sigma_{s_1.rating > s_2.rating \wedge s_2.sname = 'Bob'}(\rho_{s_1}(Sailors) \times \rho_{s_2}(Sailors)))$$

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s_1 \in Sailors (t.sid = s_1.sid \wedge \exists s_2 \in Sailors (s_2.sname = 'Bob' \wedge s_1.rating > s_2.rating))\}$$

SQL:

```

1 SELECT sid
2 FROM Sailors
3 WHERE rating > ANY (SELECT rating FROM Sailors WHERE sname = 'Bob');

```

3.3 Encontrar los sid de marineros cuyo rating es mejor que el de todos los marineros llamados Bob.

Álgebra Relacional:

$$\pi_{sid}(Sailors) - \pi_{s_1.sid}(\sigma_{s_1.rating \leq s_2.rating \wedge s_2.sname = 'Bob'}(\rho_{s_1}(Sailors) \times \rho_{s_2}(Sailors)))$$

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s_1 \in Sailors (t.sid = s_1.sid \wedge \forall s_2 \in Sailors (s_2.sname = 'Bob' \implies s_1.rating > s_2.rating))\}$$

SQL:

```

1 SELECT sid
2 FROM Sailors
3 WHERE rating > ALL (SELECT rating FROM Sailors WHERE sname = 'Bob');

```

4. División ($\div$)

4.1 Encontrar los nombres de marineros que han reservado todos los botes.

Álgebra Relacional:

$$\pi_{sname}(Sailors \bowtie (\pi_{sid,bid}(Reserves) \div \pi_{bid}(Boats)))$$

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s \in Sailors (t.sname = s.sname \wedge \forall b \in Boats (\exists r \in Reserves (r.sid = s.sid \wedge r.bid = b.bid)))\}$$

SQL (Doble NOT EXISTS):

```

1 SELECT s.sname
2 FROM Sailors s
3 WHERE NOT EXISTS (
4     SELECT b.bid FROM Boats b
5     WHERE NOT EXISTS (
6         SELECT 1 FROM Reserves r
7         WHERE r.sid = s.sid AND r.bid = b.bid
8     )
9 );

```

SQL (GROUP BY y HAVING):

```

1 SELECT s.sname
2 FROM Sailors s
3 JOIN Reserves r ON s.sid = r.sid
4 GROUP BY s.sid, s.sname
5 HAVING COUNT(DISTINCT r.bid) = (SELECT COUNT(*) FROM Boats);

```

4.2 Encontrar los nombres de marineros que han reservado todos los botes llamados BigBoat.

Álgebra Relacional:

$$\pi_{sname}(Sailors \bowtie (\pi_{sid,bid}(Reserves) \div \pi_{bid}(\sigma_{bname='BigBoat'}(Boats))))$$

SQL:

```

1 SELECT s.sname
2 FROM Sailors s
3 WHERE NOT EXISTS (
4     SELECT bid FROM Boats WHERE bname = 'BigBoat'
5     EXCEPT
6     SELECT bid FROM Reserves WHERE sid = s.sid
7 );

```

4.3 Encontrar los nombres de marineros que han reservado todos los botes que han sido reservados por marineros con menor rating que ellos.

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s_1 \in Sailors (t.sname = s_1.sname \wedge \forall s_2 \in Sailors, \forall r_2 \in Reserves ((s_2.rating < s_1.rating \wedge r_2.sid = s_2.sid \rightarrow \exists r_1 \in Reserves (r_1.sid = s_1.sid \wedge r_1.bid = r_2.bid))))\}$$

SQL:

```

1 SELECT s1.sname
2 FROM Sailors s1
3 WHERE NOT EXISTS (
4     SELECT r2.bid
5     FROM Sailors s2
6     JOIN Reserves r2 ON s2.sid = r2.sid
7     WHERE s2.rating < s1.rating
8     EXCEPT
9     SELECT r1.bid
10    FROM Reserves r1
11    WHERE r1.sid = s1.sid
12 );

```

5. Máximo / mínimo sin agregados

5.1 Encontrar los sid de marineros con el rating más alto, sin usar MAX ni LIMIT.

Álgebra Relacional:

$$\pi_{sid}(Sailors) - \pi_{s_1.sid}(\sigma_{s_1.rating < s_2.rating}(\rho_{s_1}(Sailors) \times \rho_{s_2}(Sailors)))$$

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s_1 \in Sailors (t.sid = s_1.sid \wedge \neg \exists s_2 \in Sailors (s_2.rating > s_1.rating))\}$$

SQL:

```
1 SELECT s1.sid
2 FROM Sailors s1
3 WHERE NOT EXISTS (
4     SELECT 1
5     FROM Sailors s2
6     WHERE s2.rating > s1.rating
7 );
```

5.2 Encontrar el nombre y la edad del marinero más viejo, sin usar MAX ni LIMIT.

Álgebra Relacional:

$$\pi_{sname,age}(Sailors) - \pi_{s_1.sname,s_1.age}(\sigma_{s_1.age < s_2.age}(\rho_{s_1}(Sailors) \times \rho_{s_2}(Sailors)))$$

Cálculo Relacional de Tuplas:

$$\{t \mid \exists s_1 \in Sailors (t.sname = s_1.sname \wedge t.age = s_1.age \wedge \neg \exists s_2 \in Sailors (s_2.age > s_1.age))\}$$

SQL:

```
1 SELECT s1.sname, s1.age
2 FROM Sailors s1
3 WHERE NOT EXISTS (
4     SELECT 1
5     FROM Sailors s2
6     WHERE s2.age > s1.age
7 );
```